Classroom Code- 9P2W665

Lecture-02

~~Electromagnetic~~ Electric field:

$$F = \frac{Q_1 Q_2}{4\pi \epsilon_0 R^2}$$ ϵ_0 = Permitivity of air.

$$= \frac{Q^2}{4\pi \epsilon_0 R^2}$$ | If the charge of two electrons are same.

Electric field, $E = \frac{F}{Q} = \frac{Q^2 Q}{4\pi \epsilon_0 R^2} \times \frac{1}{Q}$.

$$= \frac{Q}{4\pi \epsilon_0 R^2}$$

$$U_r = \cos\phi\, U_x + \sin\phi\, U_y$$

$$U_\theta = -\sin\phi\, U_x + \cos\phi\, U_y$$

Electron Motion in a Electric field.

Cylindrical coordinates (r, ϕ, z)

$$\left.\begin{aligned} v_r &= \cos\phi\, v_x + \sin\phi\, v_y \\ v_\phi &= -\sin\phi\, v_x + \cos\phi\, v_y \end{aligned}\right\} \text{General equation.}$$

$$\frac{d}{d\phi}(v_r) = \frac{d}{d\phi}(\cos\phi\, v_x) + \frac{d}{d\phi}(\sin\phi\, v_y)$$

$$= v_x(-\sin\phi) + v_y \cos\phi$$

$$= -\sin\phi\, v_x + \cos\phi\, v_y$$

$$= v_\phi$$

$$\frac{d}{d\phi}(v_\phi) = \frac{d}{d\phi}(-\sin\phi\, v_x) + \frac{d}{d\phi}(\cos\phi\, v_y)$$

$$= -\cos\phi\, v_x - \sin\phi\, v_y$$

$$= -(\cos\phi\, v_x + \sin\phi\, v_y)$$

$$= -v_r$$

Pointing vector calculation

$$\rho = r\,u_r + z\,u_z$$

$$\frac{d\rho}{dt} = \frac{d}{dt}(r\,u_r) + \frac{d}{dt}(z\,u_z)$$

$$= \frac{dr}{dt}u_r + r\frac{du_r}{dt} + \frac{dz}{dt}\cdot u_z + \frac{du_z}{dt}z\,.$$

$$= \frac{dr}{dt}u_r + \frac{du_r}{dt}\cdot r + \frac{dz}{dt}\cdot u_z$$

$$= u_r\frac{dr}{dt} + r\cdot\frac{du_r}{dt} + u_z\frac{dz}{dt}.$$

$\frac{du_z}{dt} = 0$.

$u_z \rightarrow$ constant

18-02-26

Helmholtz Equation

- source free (If we not apply any power region)

Maxwell equation : $\nabla \times \bar{E} = -j\omega\mu \bar{H}$ — (i)

$\nabla \times \bar{H} = j\omega\epsilon \bar{E}$ — (ii)

From (ii), $\bar{H} = \frac{j\omega\epsilon \bar{E}}{\nabla}$

Then, (i), be $\nabla \times E = -j\omega\mu \cdot \frac{j\omega\epsilon \bar{E}}{\nabla}$

$\Rightarrow \nabla \times \nabla \times \bar{E} = \omega^2 \mu\epsilon \bar{E}$ — (iii)

vector identity law,

$$\nabla \times \nabla \times A = \nabla(\nabla \cdot A) - \nabla^2 \bar{A}$$

$$\nabla \times \nabla \times \bar{E} = \nabla(\nabla \cdot E) - \nabla^2 \bar{E}$$

$$\Rightarrow \omega^2 \mu\epsilon \bar{E} = \nabla(\nabla E) - \nabla^2 \bar{E}$$

$$\Rightarrow \nabla(\nabla \cdot E) = \omega^2 \mu\epsilon \bar{E} + \nabla^2 \bar{E}$$

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For source free region, $\nabla \cdot E = 0$

$$\nabla^2 \bar{E} + \omega^2 \mu \epsilon \bar{E} = 0 \quad \text{---} \quad (4)$$

This is Helmholtz Equation for electricity of E.

If we replace the value of E the,

$$\nabla^2 \bar{H} + \omega^2 \mu \epsilon \bar{H} = 0$$

↳ Helmholtz equation for magnetic field.

Pointing theorem → (Electric field, Magnetic field) relation.

Pointing

Pointing vector for steady state is

$$\langle P \rangle = \langle E^* H \rangle = \frac{1}{2} Re(E^* H^*) \quad \text{---(1)}$$

$$P = \frac{1}{2}(E^* H^*) \quad \text{---(2)}$$

[Electric field ei]

$$\nabla \times E = -j\omega\mu H \quad \text{---(3)}$$

$$\nabla \times H = J + j\omega\epsilon E \quad \text{---(4)}$$

Dot Multiplication of (3) by H^*

$$(\nabla \times E) H^* = -j\omega\mu H \cdot H^* \quad \text{---(5)}$$

Complex conjugate of Eqn (4)

$$\nabla \times H^* = J^* - j\omega\epsilon E^* \quad \text{---(6)}$$

[Electric field এর sign change হবে complex conjugate করায়]

$\langle P \rangle = \frac{1}{2} Re(E \times H^*)$

$\nabla \times H^* = J^* - j\omega \epsilon E^*$

Dot multiplication ⑥ by E.

$\nabla \times E$

$(\nabla \times H^*) \cdot E = (J^* - j\omega \epsilon E^*) \cdot E$ — ⑦

⑦ − ⑤

$E(\nabla \times H^*) - H^*(\nabla \times E)$

$= E J^* - j\omega \epsilon \cdot E^* \cdot E + j\omega \mu H^* \cdot H$

$= E J^* - j\omega (\epsilon \cdot E^* E - \mu H^* \cdot H)$

$[E^* E = |E|^2]$

$= E J^* - j\omega (\epsilon |E|^2 - \mu |H|^2)$

$E^* E = |E|^2$

$\therefore E \cdot (\nabla \times H^*) - H^* (\nabla \times E) = E J^* - j\omega (\epsilon |E|^2 - \mu |H|^2)$

CT-01

1) Microwave is a form of electromagnetic radiation having wavelengths ranging from 30 cm to 1 mm, which corresponds to a frequency range of approximately 1 GHz to 300 GHz.

Properties of Microwave Frequencies

- Microwaves are the waves that radiate electromagnetic energy with shorter wavelength.
- Microwave are not reflected by Ionosphere
- Microwaves travel in a straight line and are reflected by the conducting surfaces
- Microwaves are easily attenuated within shorter distances
- Microwave currents can flow through a thin layer of a cable.

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1) Near field Region:

The near field is the region very close to the antenna where the electromagnetic ~~fit~~ fields are not fully formed into radiating waves and behave unpredictably.

$$NFR < \frac{2D^2}{\lambda}$$

Reactive Near field:

- Located immediately next to the antenna.
- Electric field (E) and magnetic field (H) are 90° out of phase
- Energy is mainly stored, not radiated.
- For radiation to occur, E and H must be perpendicular and in phase.

$$RNF < 0.62\sqrt{\frac{D^3}{\lambda}}$$

2. Rediative Neare field;

- Region between reactive neare field and far field.
- Fields begin to transition from reactive to radiating.
- Radiation pattern still changes with distance from the antenna

$0.62\sqrt{\frac{D^3}{\lambda}}$ ✗

$\frac{2D^2}{\lambda}$

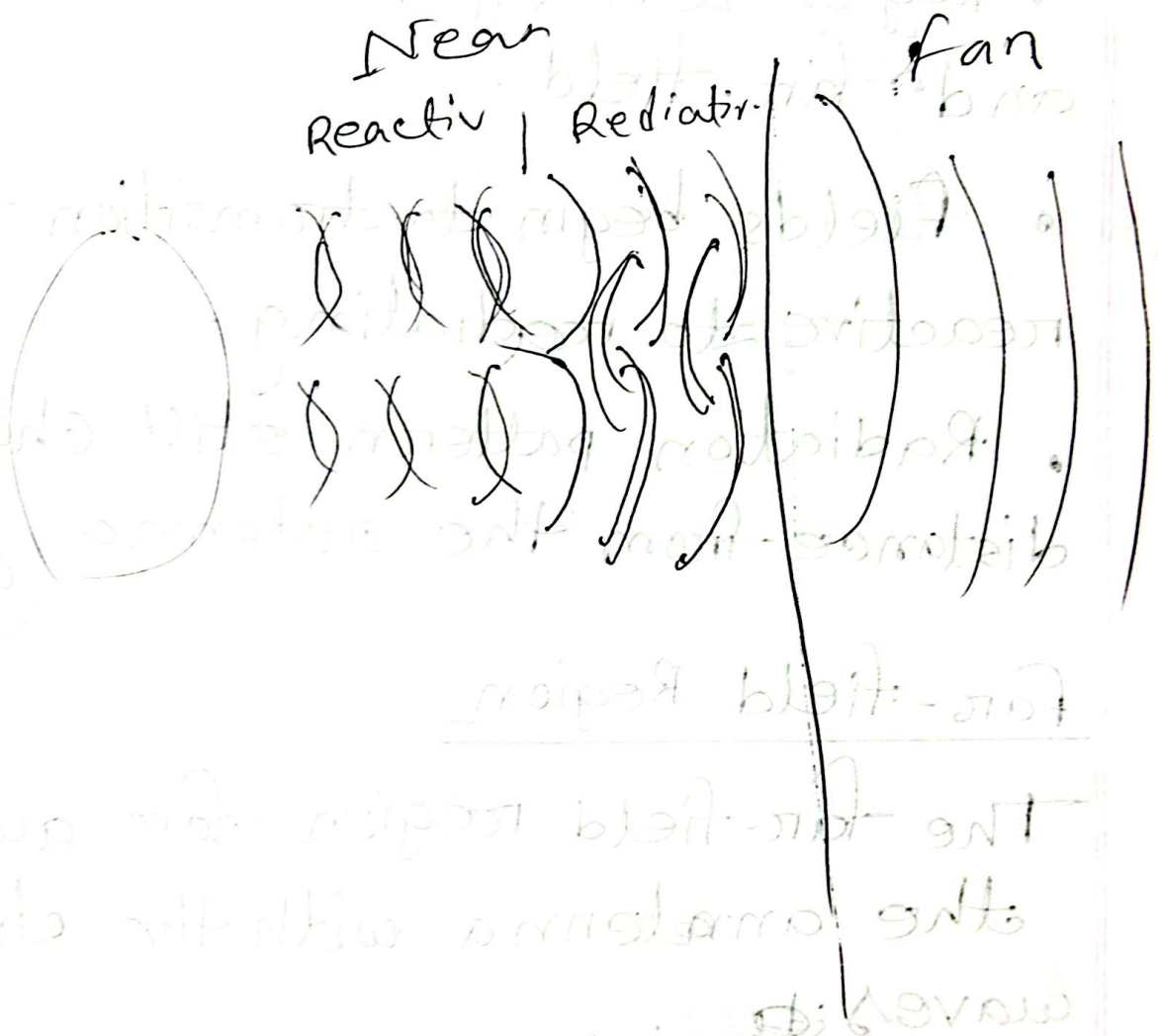
Fare-field Region

The far-field region far away from the antenna with the electromagnet waves.

- fields are dominated by radiating components.
- Electric field and magnetic field are perpendicular to each other.

$FFR > \frac{2D^2}{\lambda}$

- Radiation pattern becomes independent of distance.



$R_{FF} = \frac{2D^2}{\lambda}$

$R_{FF} = 0.62\sqrt{\frac{D^3}{\lambda}}$

LSCXku k ka
Mil
su L

Freq range in —

HF - 0.003 - 0.030
VHF - 0.030 - 0.300
UHF - 0.300 - 1.00
L - 1 - 2
S - 2 - 4
C - 4 - 8
X - 8 - 12
ku - 12 - 18
k - 18 - 27
ka - 27 - 90
Milli - 90 - 300
Sub - > 300

0.003
0.030.
0.3

(47/28)

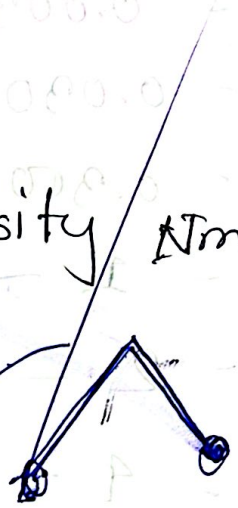
04-03-2025

Plane wave propagation in free space.

Sky wave

maximum electron density Nmax.

$F_{cr} = 9\sqrt{N_{max}}$ Hz.



F_{cr} → Critical frequency: minimum frequency for reflected back to the earth

Wavelength shorter than about 4m (f = 74.9 MHz).

Ground Wave

Maximum electron density N_{max}.

$F_{cr} = 9\sqrt{N_{max}}$

Plane-wave propagation in Free space.

The power density,

$$P_d = \frac{P_t g_t}{4\pi R^2}$$

(P_t: Transmitted power, g_t: transmitting antenna Gain of transmitter.)

$$P_r = P_d \times A_e = \left(\frac{P_t g_t}{4\pi R^2}\right)\left(\frac{\lambda^2}{4\pi} g_r\right)$$

(g_r: receiving antenna gain.)

$$= 10\log\left(\frac{P_t g_t \lambda^2 g_r}{16\pi^2 R^2}\right)$$

$$= 10\log(P_t g_t \lambda^2 g_r) - 20\log(4\pi R)$$

$$= 10\log P_t + 10\log g_t + 10\log g_r + 20\log\lambda - 20\log 4\pi R$$

$$= 10\log P_t + 10\log g_t + 10\log g_r - 20(\log 4\pi R - \log\lambda)$$

$$= 10\log P_t + 10\log g_t + 10\log g_r - 20\log\frac{4\pi R}{\lambda}$$

$$= P_t + G_t + G_r - 20\log\frac{4\pi R}{\lambda} \text{ dBW}$$

loss

free space attenuation in decibels.

Plane-Wave Propagation in Lossless Dielectric

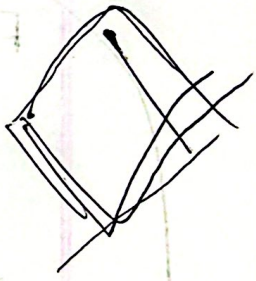
$$\eta = \sqrt{\frac{\mu}{\epsilon}} = \sqrt{\frac{\mu_0}{\epsilon_r \epsilon_0}} = \frac{377}{\sqrt{\epsilon_r}} \text{ ohms}$$

$$\sqrt{\frac{\mu_0}{\epsilon_0}} = 377$$

Attenuation constant α is zero, and the phase constant β.

$$\beta = \omega\sqrt{\mu\epsilon}$$

$$\sqrt{\frac{\mu_0}{\epsilon_r \epsilon_0}} = \frac{377}{\sqrt{\epsilon_r}}$$



$$\mu = \eta = \sqrt{\frac{\mu}{\epsilon}} = \sqrt{\frac{\mu_0}{\epsilon_r \epsilon_0}} = \frac{377}{\sqrt{\epsilon_r}}$$

Attenuation constant $\alpha = 0$ and the phase constant

$$\gamma = \omega\sqrt{\mu\epsilon}$$

From complex frequency solution:

$$\gamma = \sqrt{j\omega\mu(\sigma + j\omega\epsilon)}$$

$$= \sqrt{j\omega\mu \cdot j\omega\epsilon\left(1 - \frac{j\sigma}{\omega\epsilon}\right)}$$

$$= j\omega\sqrt{\mu\epsilon\left(1 - \frac{j\sigma}{\omega\epsilon}\right)}$$

$$= j\omega\sqrt{\mu\epsilon\left(-\frac{j\sigma}{\omega\epsilon}\right)}$$

$$\sigma + j\omega\epsilon$$

$$= j\omega\epsilon + \sigma$$

$$= j\omega\epsilon\left(1 + \frac{\sigma}{j\omega\epsilon}\right)$$

$$= j\omega\epsilon\left(1 + \frac{j\cdot\sigma}{j^2\omega\epsilon}\right)$$

$$= j\omega\epsilon\left(1 - \frac{j\sigma}{\omega\epsilon}\right)$$

For good conducton:

$$\sigma \gg \omega\epsilon$$

$$\Rightarrow \frac{\sigma}{\omega\epsilon} \gg 1, \quad 1 - \frac{j\sigma}{\omega\epsilon}$$

$$\therefore 1 - \frac{j\sigma}{\omega\epsilon} \simeq \left(-\frac{j\sigma}{\omega\epsilon}\right)$$

$$\gamma = j\omega\sqrt{\mu\epsilon}\cdot\sqrt{\frac{\sigma}{\omega\epsilon}}\cdot\sqrt{(-j)}$$

$$= j\cdot\sqrt{\omega}\cdot\sqrt{\omega}\cdot\sqrt{\mu\epsilon}\cdot\frac{\sqrt{\sigma}}{\sqrt{\omega}\cdot\sqrt{\epsilon}}\sqrt{(-j)}$$

$$= j\sqrt{\omega\mu\epsilon}\cdot\sqrt{\frac{\sigma}{\epsilon}}\cdot\sqrt{(-j)}$$

Excluding constants:

$$\gamma = j\sqrt{\omega\mu\epsilon}\cdot\sqrt{(-j)}$$

$$= j\sqrt{\omega\mu\epsilon}\cdot\left(\frac{1-j}{\sqrt{2}}\right)$$

$$= \sqrt{\omega\mu\epsilon}\cdot\left[\frac{j-j^2}{\sqrt{2}}\right]$$

$$= \sqrt{\omega\mu\epsilon}\cdot\frac{1+j}{\sqrt{2}}$$

$$= (1+j)\cdot\sqrt{\frac{\omega\mu\epsilon}{2}}$$

$$= (1+j)\sqrt{\frac{2\pi f\mu\epsilon}{2}}$$

$$= (1+j)\sqrt{\pi f\mu\epsilon}$$ → Permittivity

From complex variable →

$$\sqrt{(-j)} = \left(\frac{1-j}{\sqrt{2}}\right)$$

$$= \frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}}$$

Relationship with conductivity

$$\gamma = j\sqrt{\omega\mu\epsilon}\cdot\sqrt{\frac{\sigma}{\epsilon}}\cdot\sqrt{(-j)}$$

$$= j\sqrt{\frac{\omega\mu\epsilon\sigma}{\epsilon}}\cdot\sqrt{(-j)}$$

$$= \sqrt{\omega\mu\sigma}\cdot j\left(\frac{1-j}{\sqrt{2}}\right)$$

$$= \sqrt{\omega\mu\sigma}\cdot\frac{1+j}{\sqrt{2}}$$

$$= \sqrt{\frac{\omega\mu\sigma}{2}}\cdot(1+j) = \sqrt{\frac{2\pi f\mu\sigma}{2}}(1+j)$$

$$= \sqrt{\pi f\mu\sigma}\cdot(1+j).$$

Good conducton →

$$\alpha = \beta$$

★★★ Lossy Dielectric :

Derive the plane-wave propagation for good conductor

Lossy conduction

$$\gamma = \sqrt{j\omega\mu(\sigma + j\omega\epsilon)}$$

$$= \sqrt{j\omega\mu \cdot j\omega\epsilon\left(-\frac{j\sigma}{\omega\epsilon}\right)} \cdot j\omega\epsilon\left(1 + \frac{\sigma}{j\omega\epsilon}\right)$$

$$= j\omega\sqrt{\mu\epsilon\left(-\frac{j\sigma}{\omega\epsilon}\right)} \; j\omega\epsilon\left(1 - \frac{j\sigma}{\omega\epsilon}\right)$$

$$\gamma = j\omega\sqrt{\mu\epsilon}\left(1 - \frac{j\sigma}{\omega\epsilon}\right)^{\frac{1}{2}}$$

$$= j\omega\sqrt{\mu\epsilon} \cdot \sqrt{\frac{\omega\epsilon}{\omega\epsilon}} \cdot \sqrt{(-j)}$$

$$= j \cdot \sqrt{\omega} \cdot \sqrt{\omega} \cdot \sqrt{\mu\epsilon} \cdot \sqrt{\frac{\sigma}{\omega\epsilon}} \cdot \sqrt{(-j)}$$

$$= j\sqrt{\omega\mu\epsilon}\left(\sqrt{-j}\right)$$

Plane-wave in Good conductors.

→ From complex frequency solution:

$$\gamma = \sqrt{j\omega\mu(\sigma + j\omega\epsilon)}$$

$$= \sqrt{j\omega\mu \cdot j\omega\epsilon\left(1 - \frac{j\sigma}{\omega\epsilon}\right)}$$

$$= j\omega\sqrt{\mu\epsilon\left(-\frac{j\sigma}{\omega\epsilon}\right)}.$$

$$\sigma + j\omega\epsilon$$
$$= j\omega\epsilon\left(1 + \frac{\sigma}{j\omega\epsilon}\right)$$
$$= j\omega\epsilon\left(1 + \frac{j\sigma}{j^2\omega\epsilon}\right)$$
$$= j\omega\epsilon\left(1 - \frac{j\sigma}{\omega\epsilon}\right).$$

For good conductors:

$$\sigma \gg \omega\epsilon$$

$$\therefore \frac{\sigma}{\omega\epsilon} \gg 1 \quad \therefore$$

$$\therefore 1 - \frac{j\sigma}{\omega\epsilon} \approx \left(-\frac{j\sigma}{\omega\epsilon}\right)$$

$$\gamma = j\omega\sqrt{\mu\epsilon} \cdot \sqrt{\frac{\sigma}{\omega\epsilon}} \cdot \sqrt{(-j)}$$

$$= j \cdot \sqrt{\omega} \cdot \sqrt{\omega} \cdot \sqrt{\mu\epsilon} \cdot \frac{\sqrt{\sigma}}{\sqrt{\omega} \cdot \sqrt{\epsilon}} \cdot \sqrt{(-j)}$$

$$= j\sqrt{\omega\mu\epsilon} \cdot \sqrt{\frac{\sigma}{\epsilon}} \cdot \sqrt{(-j)}.$$

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· Excluding constants:

$$\gamma = j\sqrt{\omega\mu\epsilon}\cdot\sqrt{(-j)}$$

$$= j\sqrt{\omega\mu\epsilon}\cdot\left(\frac{1-j}{\sqrt{2}}\right)$$

$$= \sqrt{\omega\mu\epsilon}\left(\frac{j-j^2}{\sqrt{2}}\right)$$

$$= \sqrt{\omega\mu\epsilon}\cdot\frac{1+j}{\sqrt{2}}$$

$$= (1+j)\cdot\sqrt{\frac{\omega\mu\epsilon}{2}}$$

$$= (1+j)\sqrt{\frac{2\pi f\mu\epsilon}{2}}$$

$$= (1+j)\sqrt{\pi f\mu\epsilon}.$$

$$\sqrt{(-j)} = \frac{1-j}{\sqrt{2}}$$

Relationship with Permitivity.

$$\gamma = j\sqrt{\omega\mu\epsilon}\cdot\sqrt{\frac{\sigma}{\epsilon}}\cdot\sqrt{(-j)}$$

$$= j\cdot\sqrt{\frac{\omega\mu\epsilon\sigma}{\epsilon}}\cdot\sqrt{(-j)}$$

$$= \sqrt{\omega\mu\sigma}\cdot j\cdot\frac{1-j}{\sqrt{2}}$$

$$= \sqrt{\omega\mu\sigma}\cdot\frac{1+j}{\sqrt{2}}.$$

$$= \sqrt{\frac{\omega\mu\sigma}{2}} \times (1+j)$$

$$= (1+j)\sqrt{\pi f \mu \sigma}$$

Relationship with conductivity

Lossy Dielectric

$$\gamma = \sqrt{j\omega\mu(\sigma + j\omega\epsilon)}$$

$$= \sqrt{j\omega\mu \cdot j\omega\epsilon\left(1 - \frac{j\sigma}{\omega\epsilon}\right)}$$

$$= j\omega\sqrt{\mu\epsilon\left(1 - \frac{j\sigma}{\omega\epsilon}\right)}$$

$$= j\omega\sqrt{\mu\epsilon}\left(1 - \frac{j\sigma}{2\omega\epsilon}\right)$$

$$\gamma = \alpha + j\beta$$

$$\alpha = \frac{j\omega\sqrt{\mu\epsilon}\times\left(-\frac{j\sigma}{2\omega\epsilon}\right)}{2}$$

$$= \frac{\sigma}{2}\sqrt{\frac{\mu}{\epsilon}}$$

$$\beta = \omega\sqrt{\mu\epsilon}$$

$$j\omega\sqrt{\mu\epsilon}$$

Binomial Expansion

$$\sqrt{1 - \frac{j\sigma}{\omega\epsilon}}^{\left(\frac{1}{2}\right)} \approx$$

$$1 - \frac{j\sigma}{2\omega\epsilon}$$

$$(1+x)^n = 1 + nx$$

$$n = \frac{1}{2}$$

$$\sqrt{1 - \frac{j\sigma}{\omega\epsilon}} = \left(1 - \frac{j\sigma}{\omega\epsilon}\right)^{\frac{1}{2}}$$

$$1 + \left(-\frac{j\sigma}{2\omega\epsilon}\right)$$

$$1 - \frac{j\sigma}{2\omega}$$

* Plane wave prop in free space

the powers density,

$$P_d = \frac{P_t g_t}{4\pi R^2} \quad -1$$

Where, P_t = transmitted p

$$P_{rt} = P_d \times A_e$$

$$= \left(\frac{P_t g_t}{4\pi R^2}\right) \times \left(\frac{\lambda^2}{4\pi} g_{rt}\right)$$

$$= \frac{P_t g_t \lambda^2 g_r}{(4\pi R^2)^2}$$

$$= 10 \log (P_t g_t \lambda^2 g_r) - 20 \log (4\pi R)$$

$$= 10 \log P_t + 10 \log g_t + 20 \log \lambda + 10 \log g g_r - 20 \log (4\pi R)$$